

Espresso Preparation Guide

This guide covers how to prepare and dial in espresso, organized into three stages: coffee selection, dialing in the shot, and pulling and tasting the shot. A quick-reference table of roast-specific parameters is included at the end.

1. Coffee Selection

1.1 Choosing coffee for espresso

Specialty coffee produces more predictable espresso results. Specialty coffee lists detailed information on the bag, such as origin, roast date, processing method, and variety. If the coffee is not specialty grade, switching to specialty coffee is recommended for more consistent results.

1.2 Identifying the roast degree

Espresso beans are roasted to one of three degrees. The roast degree can be identified by bean color and by a crush test (crushing a bean to feel how easily it breaks).

- **Light roast:** lighter in color and harder to crush.
- **Medium roast:** mid-brown color and slightly easier to crush.
- **Dark roast:** dark brown to black color and very easy to crush.

Visually inspect the beans and perform the crush test to confirm the roast level before brewing.

1.3 Checking the roast date (resting time)

Coffee should rest for an optimal window after roasting before being used for espresso. Coffee that is too fresh has excess CO₂ buildup; coffee that is too old goes stale. Optimal resting time after the roast date, by roast degree:

- **Dark roast:** 3 to 7 days after the roast date.
- **Medium roast:** 7 to 14 days after the roast date.
- **Light roast:** 14 to 28 days after the roast date.

1.4 Checking the processing method

The processing method is listed on the coffee bag, for example washed, natural, or other. Complex or heavily processed coffees (for example, thermal shock or double fermented) tend to extract more quickly. Adjust extraction time and brew ratio based on the processing method.

2. Dialing In Espresso

2.1 Setting the dose

The dose is the weight of ground coffee used for a shot (for example, 18 g). The dose should match the basket size. Avoid overdosing: if the basket is overfilled, the shower screen touches the coffee puck, which harms extraction.

2.2 Setting the brew ratio (water to coffee)

The brew ratio is the ratio of dry coffee dose to liquid espresso output. Choose the ratio based on roast degree:

- **Light roast:** 1:2.5 to 1:3.
- **Medium roast:** 1:2.
- **Dark roast:** 1:1.5 to 1:1.

The chosen ratio should match the coffee's roast degree and processing method.

2.3 Setting the grind size

Set the grind size based on roast degree, then fine-tune it by observing the shot's flow rate.

Starting point by roast degree:

- **Light roast:** grind finer.
- **Dark roast:** grind coarser.

Fine-tuning by flow rate:

- If the shot flows too fast, grind finer.
- If the shot flows too slow, grind coarser.

2.4 Setting the water temperature

Set the brew water temperature based on roast degree:

- **Dark roast:** 80 to 85 °C.
- **Medium roast:** 85 to 90 °C.
- **Light roast:** 90 °C or above.

If the machine's temperature control is not accurate, use temperature surfing: flush the group head to adjust the brew temperature before pulling the shot.

3. Pulling the Shot

3.1 Puck preparation

Prepare the coffee puck before pulling the shot:

- Distribute the grounds evenly in the basket.
- Tamp with consistent pressure.
- If using a regular tamper, make sure it sits snug and level.
- Optional: use the Ross Droplet Technique (RDT), adding a tiny amount of water to the beans before grinding, to reduce static and grind retention.

3.2 Tracking brew time

Start the timer as soon as the pump engages. A typical target brew time is 25 to 30 seconds.

- If the shot pulls too fast, grind finer or increase the dose.
- If the shot pulls too slow, grind coarser or decrease the dose.

3.3 Tasting and adjusting

Taste the shot and adjust based on flavor:

- **Too sour or salty:** grind finer or pull the shot longer.
- **Too bitter or dry:** grind coarser or pull the shot shorter.
- **Balanced:** the espresso is dialed in.

Quick Reference: Parameters by Roast Degree

Parameter	Light roast	Medium roast	Dark roast
Bean color	Lighter	Mid-brown	Dark brown to black
Crush test	Harder to crush	Slightly easier to crush	Very easy to crush
Resting time after roast	14 to 28 days	7 to 14 days	3 to 7 days
Brew ratio (coffee:water)	1:2.5 to 1:3	1:2	1:1.5 to 1:1
Grind size	Finer	Medium	Coarser
Water temperature	90 °C or above	85 to 90 °C	80 to 85 °C